Part 1 ClinVar labels added after the release used by the AVI developers (2025-06-15)

ClinVar, October 2026 release

1,762,164 variants classified P/LP or B/LB (GRCh38)

Excluded

1,426,176 already P/LP or B/LB at or before that release
(ClinVar 2025-06-15 VCF; 2025-07 and 2024-07 releases)
123,767 at AVI developers' benchmark positions or from a development submitter
818 older record of unresolved identity at the same position
17 identity collisions

New-label set

211,386 variants

Not in the main set (filters in this order)

31,469 not SNV
46,338 coding other than synonymous (missense, nonsense, other)
482 HGVS name not parsed
249 no review star
668 no MANE Select name
40,403 on training chromosomes (supporting analysis)
4,111 not rare in gnomAD v4.1 (FAF95 ≥ 0.001) (supporting analysis)

Main set

87,666 variants: SNV; intronic, UTR or synonymous; ≥1 review star; MANE Select name; untouched and validation chromosomes; rare in gnomAD v4.1

Main comparison

intronic ≥3 bp from an exon: 449 P/LP, 28,188 B/LB

Calibration sample (thresholds derived here)

labelled variants outside the new-label set, non-canonical intronic and UTR, ≥1 star: all 4,001 P/LP and 40,000 B/LB drawn (stratified by region and chromosome group); rare and scored by AVI: 3,248 P/LP, 23,327 B/LB. Thresholds tested on the main set.

Part 2 Experimental splicing measurements (no predictor selected the variants)

Six main datasets (pooled by random effects); comparison sets reported separately

MST1R exon 11 (minigene): 1,743 (774 splice-altering)
COMPASS (minigene libraries): 31,975 (7,063 splice-altering)
CD19 (minigene): 3,588 (178 splice-altering)
POU1F1 exon 2 (minigene): 1,034 (96 splice-altering)
WT1 exon 9 (minigene): 485 (47 splice-altering)
OpenSplice (MPRA, 599 exons): 295,405 (165,549 splice-altering)

Comparison: MFASS: 27,127 (1,039 splice-altering); OpenSplice at AVI benchmark positions: 13,993 (5,936 splice-altering).
Alleles of the Part 1 new-label set and at AVI developers' benchmark positions were removed from the main datasets.

Tools: AVI against SpliceAI (raw and masked), Pangolin, CADD v1.7 and conservation (phyloP 100-way; AVI's inputs phastCons 470-way and phyloP 241-way).
Added after the main results: SpliceAI2.